

Two Case Studies in Natural Language Processing

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1st task : Multiclass classification for cultural items

Cultural agnostic



Cultural Representative



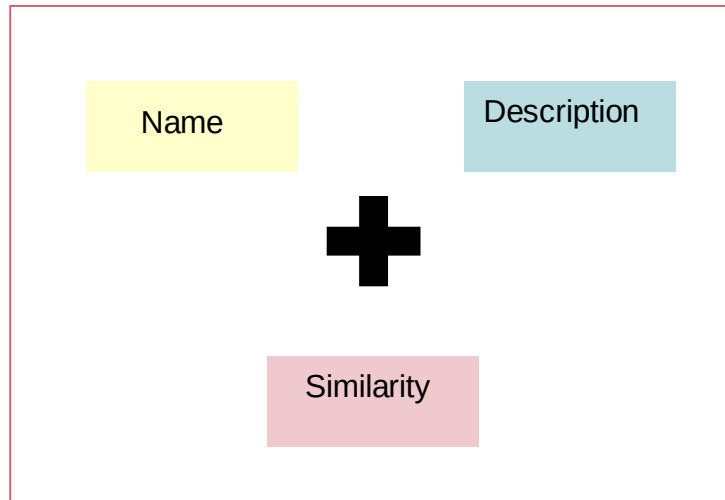
Cultural Exclusive



Item	Name	Description	Type	Category	Subcategory	Label
Link to wikidata	Name of the item	Small text with item description	Entity or concept	Like sports, politics, ecc.	Like sports equipment, law, ecc.	Cultural agnostic, representative or exclusive

1st task: Approaches

Non-LLM Approach

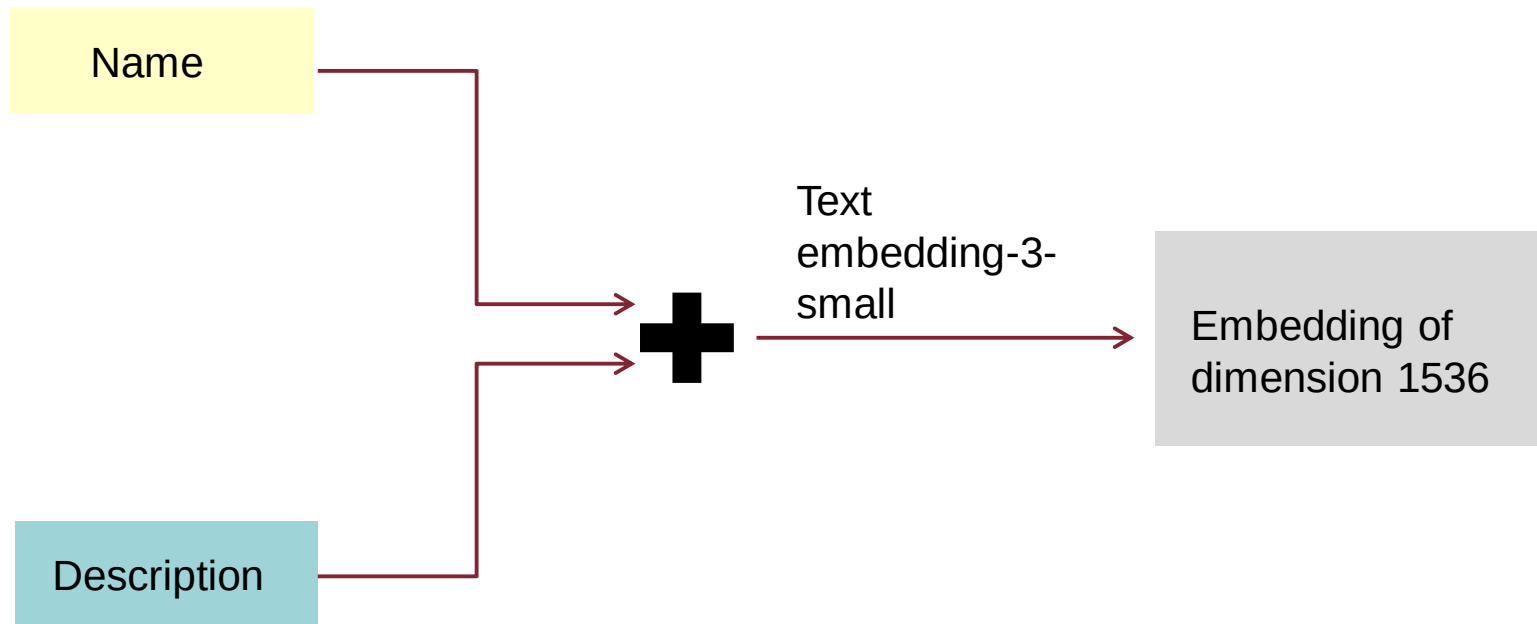


LLM Approach

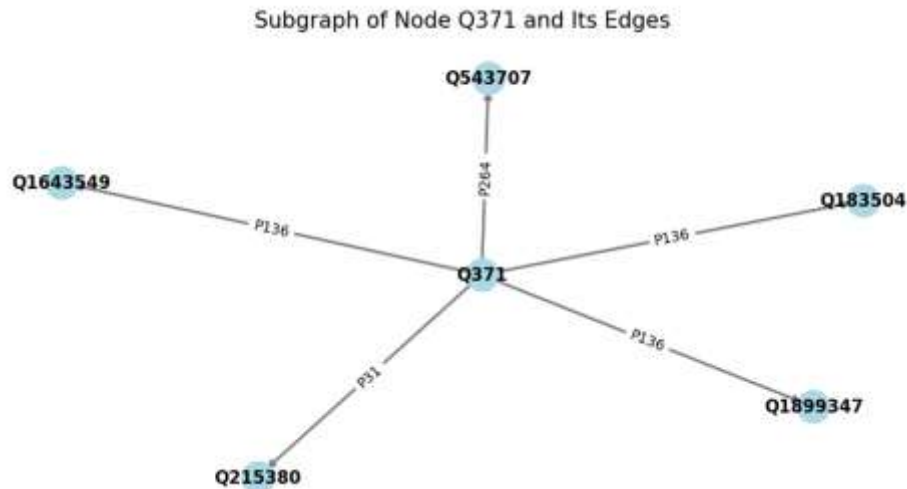
«You are a classification expert. Based on the information below, classify the item into the correct category:
[LANG]...[NAME]...[DESC]...[TYPE]...»

xlm-roberta-base

1st Approach: Non-LLM



1st Approach: Non-LLM



Each item in the dataset is a node.

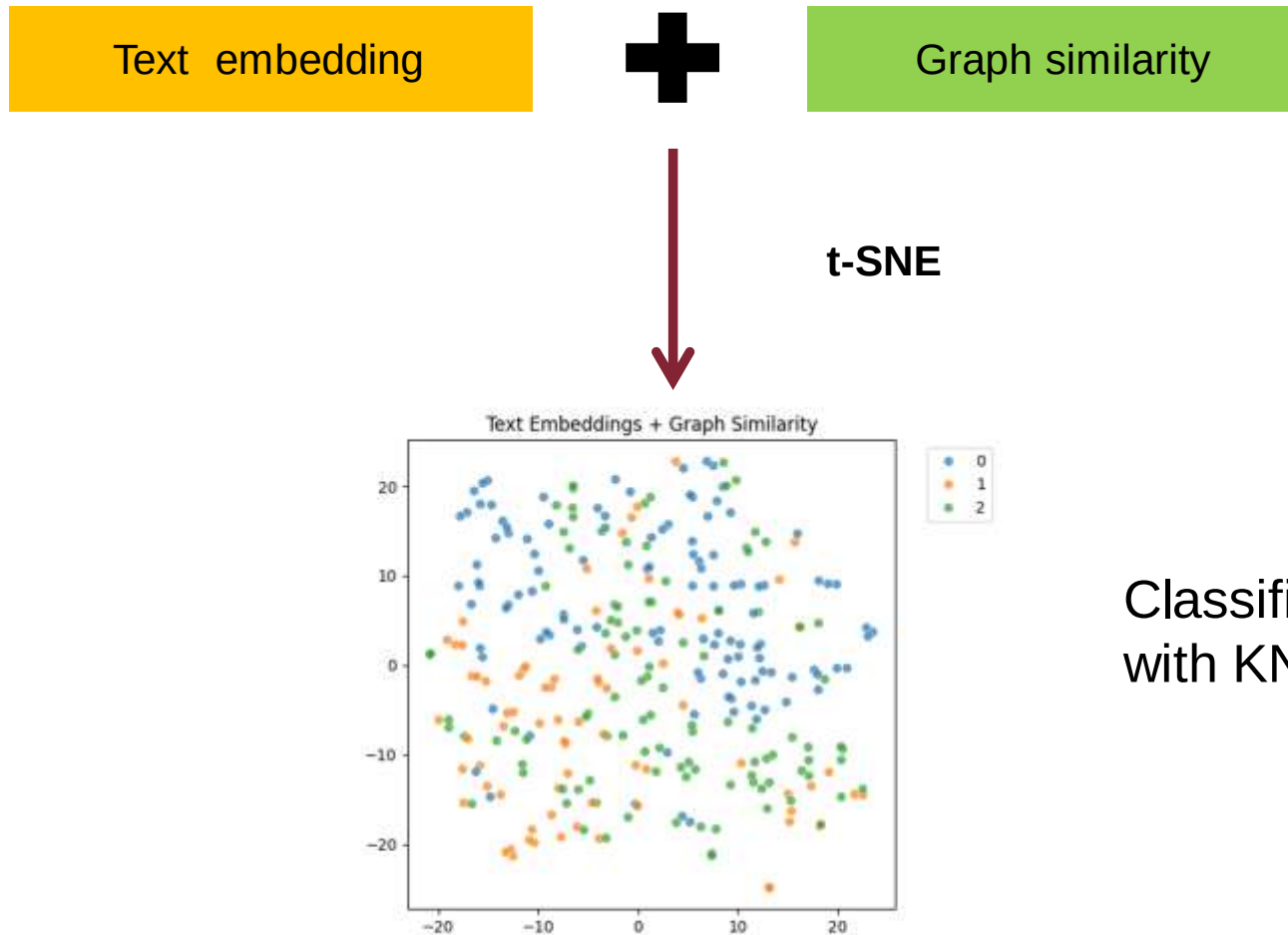
For each item we found up to 10 neighbors using a SPARQL query

For each couple Q1, Q2:

$$\text{similarity}(Q1, Q2) = \frac{|N_{Q1} \cap N_{Q2}|}{|N_{Q1} \cup N_{Q2}|}$$

For each item we collected the mean similarity and the max similarity

1st Approach: Non-LLM



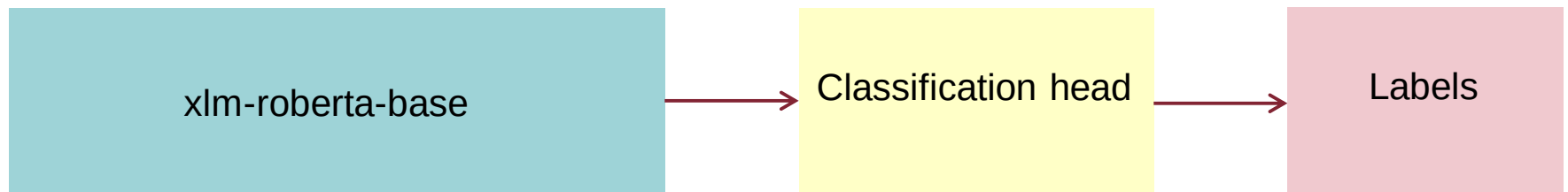
2nd approach: LLM

«You are a classification expert. Based on the information below, classify the item into the correct category: [LANG]...[NAME]...[DESC]...[TYPE]...»



Original language of wikidata item, found using a SPARQL query to Wikidata endpoint.

Fine-tuning of 10 epochs with R-Drop Loss



1st task: Results

Non-LLM method

Class	Precision	Recall	F1-score
CA	0.89	0.93	0.91
CE	0.59	0.75	0.66
CR	0.76	0.57	0.65
Acc			0.76
Macro	0.75	0.75	0.74
Weighted	0.77	0.76	0.75

LLM method

Class	Precision	Recall	F1-score
CA	0.89	0.95	0.92
CE	0.75	0.62	0.68
CR	0.72	0.76	0.74
Acc	-	-	0.80
Macro	0.79	0.77	0.78
Weighted	0.79	0.80	0.79

2nd Task: Sentence Splitting

Divide a text into sentences by recognize the [EoS] tokens.

A fundamental pre processing step in NLP

Dataset used is from 'I promessi Sposi' by Alessandro Manzoni

2 Encoder-only models

2 Small Generative models



1st Encoder-only model

Pre-trained encoder: bert-base-italian-cased

Chunking: 64 tokens + padding

Fine-tuning for 20 epochs using Weighted Cross Entropy

Regularization with Dropout

First 4 layers are freezed



2nd Encoder-only model

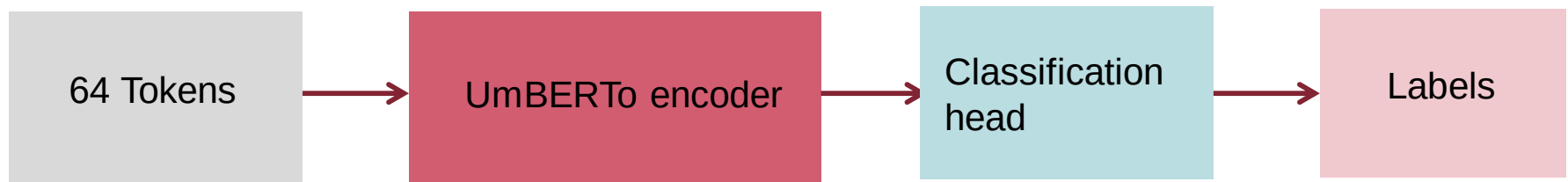
Pre-trained encoder: umberto-commoncrawl-cased-v1

Chunking: 64 tokens + padding

Fine-tuning for 20 epochs using Focal Loss

Regularization with Dropout

First 3 layers are freezed



1st Generative Approach Model

Base Model used: meta-llama/Llama-3.2-1B

Training: LoRA fine-tuning for 8 epochs

'Split the text with [EoS] tokens:\n' + 8 tokens



Llama – 3.2 – 1B



Input text with [EoS]

2nd Generative Approach Model

Base Model used: google/gemma-1.1-2b-it

Training: LoRA fine-tuning for 7 epochs

"Segna con [EoS] ogni fine frase.\n Testo: " + 16 tokens + "\n Output: "



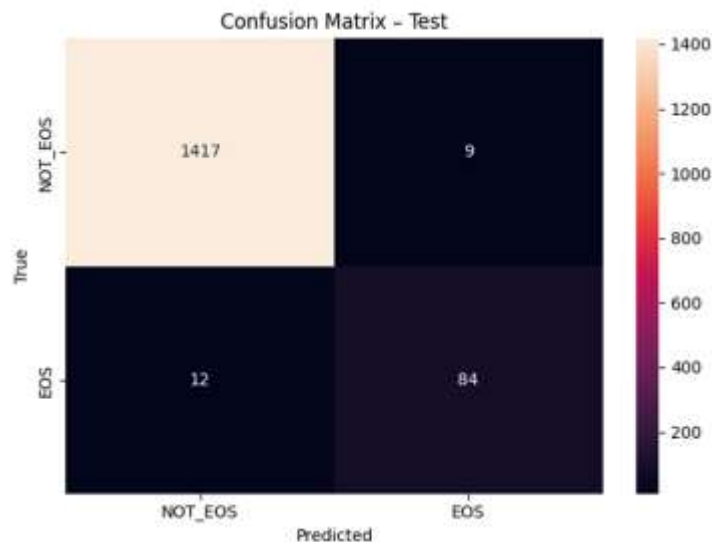
gemma-1.1-2b-it



Input text with [EoS]

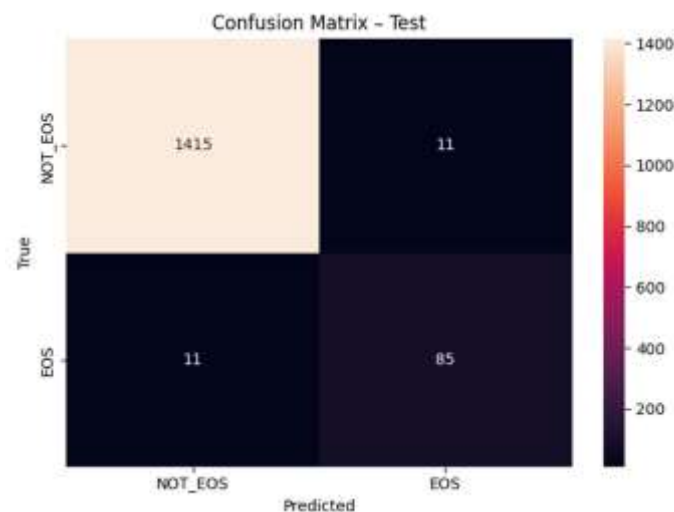
2nd Task - Results : 1st Approach

BERT



Class	Precision	Recall	F1-score
NOT_EOS	0.99	0.99	0.99
EOS	0.90	0.88	0.89
Acc	0.99		
Macro	0.95	0.93	0.94
Weighted	0.99	0.99	0.99

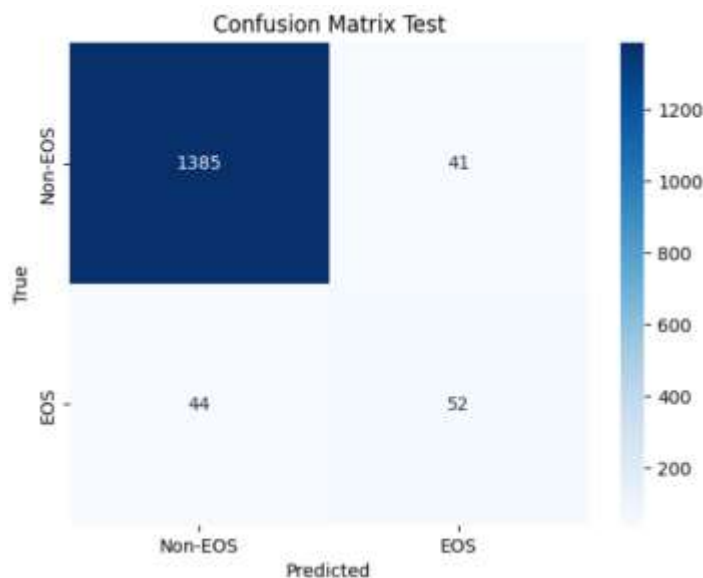
UmBERTo



Class	Precision	Recall	F1-score
NOT_EOS	0.99	0.99	0.99
EOS	0.89	0.89	0.89
Acc	0.99		
Macro	0.94	0.94	0.94
Weighted	0.99	0.99	0.99

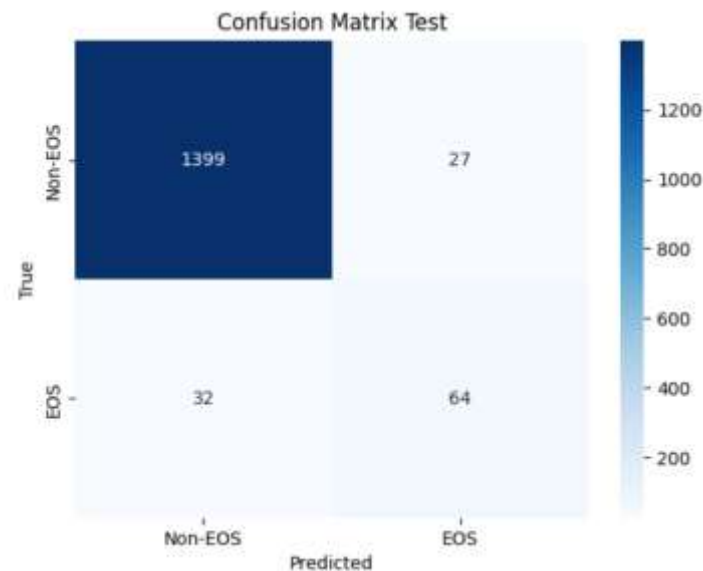
2nd Task - Results : 2nd Approach

Llama



Class	Precision	Recall	F1-score
NOT_EOS	0.97	0.97	0.97
EOS	0.56	0.54	0.55
accuracy			0.94
marco avg	0.76	0.76	0.76
weighted avg	0.94	0.94	0.94

Gemma



Class	Precision	Recall	F1-score
NOT_EOS	0.98	0.98	0.98
EOS	0.70	0.67	0.68
accuracy			0.96
marco avg	0.84	0.82	0.83
weighted avg	0.96	0.96	0.96

Thank you for your attention!